ChatGPT gpt-6

Here's the TenderOps workflow diagram:

If you want, I can:

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2026-10-10 13:19:35 You

Can u summarize the idea in detail

ChatGPT gpt-6

TenderOps: A Detailed Product & Technical Overview

1. The core idea

TenderOps is an AI-powered tender intelligence and bid preparation platform that helps companies understand government and enterprise tenders, assess whether they qualify, gather supporting evidence, and prepare a complete, compliance-checked bid package.

Tendering is often a tedious, document-heavy process. A company might receive a tender containing hundreds of pages of eligibility conditions, technical specifications, financial requirements, submission instructions, declarations, and annexures. The company must then compare those requirements against its own credentials, certificates, financial statements, past projects, and other documentation.

Missing even one mandatory condition can disqualify an otherwise strong bid.

TenderOps aims to automate much of this process using a coordinated system of AI agents, while retaining human oversight for important decisions and approvals.

<AsyncImageGroup query={["government procurement tender documents paperwork desk","business professionals reviewing contract documents meeting","enterprise document compliance checklist laptop"]} aspectRatio="5:4" layout="bento" size="lg"/>

The product is not simply a chatbot that summarizes PDFs. It is intended to be an **end-to-end tender execution workflow** that takes a company from raw tender documents to an evidence-backed, approval-ready bid package.

Its central principle is:

> Understand the tender, prove compliance using evidence, prepare the bid, independently verify it, and escalate anything that cannot be safely resolved.

2. The problem TenderOps solves

The current tender preparation process typically involves several interconnected tasks.

- **Tender analysis:** Reading long documents to identify eligibility conditions, mandatory requirements, deadlines, technical specifications, and submission instructions.
- **Eligibility assessment:** Determining whether the company meets minimum turnover, experience, certification, technical capability, and other requirements.
- **Evidence collection:** Finding the relevant documents from a company's existing repository and matching them to individual tender requirements.
- **Compliance management:** Identifying missing certificates, inconsistent figures, expired documents, and unsupported claims.
- **Bid preparation:** Filling forms, drafting declarations, assembling annexures, and preparing technical or commercial responses.
- **Quality assurance:** Checking the completed bid against the original tender to catch omissions before submission.

These tasks are often distributed across business development, finance, legal, engineering, and compliance teams. They involve repetitive work, frequent back-and-forth, and substantial risk from human error.

TenderOps brings these activities into a single orchestrated workflow.

The intended transformation

Input: Unstructured documents

```
text

  <text color="secondary" size="sm">Tender PDFs, company credentials, financial records,
certificates, previous bids, and technical documents.</text>
</box>
```

Processing: Coordinated AI workflow

```
text
```

```
  <text color="secondary" size="sm">Requirement extraction, evidence matching, eligibility
checks, document generation, validation, and recovery.</text>
</box>
```

Output: Approval-ready bid package

```
text

  <text color="secondary" size="sm">Compliance matrix, evidence references, completed draft
documents, identified blockers, and human approval status.</text>
</box>
```

The important distinction is that the system should not merely produce plausible answers. It should establish *why* a requirement is considered satisfied, which document supports that conclusion, and what remains unresolved.

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3. Who would use TenderOps?

The initial target customers could be companies that regularly respond to government or enterprise procurement opportunities.

Infrastructure and engineering contractors

```
text

  <text color="secondary" size="sm">Companies bidding for construction, electrical, civil
engineering, and public infrastructure projects.</text>
</box>
```

IT services and system integrators

```
text

  <text color="secondary" size="sm">Businesses responding to software, networking,
cybersecurity, cloud, and digital transformation tenders.</text>
</box>
```

Manufacturers and suppliers

```
text

  <text color="secondary" size="sm">Businesses that repeatedly submit product specifications,
quality certificates, financial credentials, and supplier declarations.</text>
</box>
```

For an initial product, a narrow industry would be preferable to supporting every kind of tender. A relatively standardized tender category would make it easier to build reliable extraction rules, validation checks, and document templates.

4. The complete TenderOps workflow

The workflow is organized into ten major stages, with an orchestrator coordinating the agents and deciding what happens next.

Stage 1 — Document upload

```
text

  <text color="secondary" size="sm">The user uploads the tender and relevant company documents.
The system creates a tender workspace and associates each file with its source and metadata.
</text>
</box>
```

Stage 2 — Document ingestion

```
text

  <text color="secondary" size="sm">PDFs and other supported formats are parsed to extract
text, tables, page numbers, headings, and metadata. Scanned files may require OCR. Source
references are preserved for later verification.</text>
</box>
```

Stage 3 — Tender Analyst Agent

```
text

  <text color="secondary" size="sm">The agent turns the tender into structured requirements:
eligibility criteria, mandatory documents, deadlines, deliverables, technical specifications,
commercial conditions, and submission rules.</text>
</box>
```

Stage 4 — Orchestration and persistent state

```
text

  <text color="secondary" size="sm">The orchestrator creates tasks, routes them to the
appropriate agents, tracks completed and pending work, and saves checkpoints so interrupted
workflows can resume.</text>
</box>
```

Stage 5 — Evidence Agent

```
text
  <text color="secondary" size="sm">The agent searches the company's document repository for
relevant certificates, past-project records, financial statements, and other supporting
evidence. It maps each requirement to the evidence available.</text>
</box>
```

Stage 6 — Compliance and eligibility checks

```
text

  <text color="secondary" size="sm">Requirements are compared against the available evidence.
The system identifies satisfied conditions, missing evidence, potential disqualifiers,
inconsistencies, and ambiguous cases.</text>
</box>
```

Stage 7 — Recovery or bid preparation

```
text

  <text color="secondary" size="sm">If information is missing or contradictory, the workflow
flags the blocker, saves its state, and requests human input when necessary. Otherwise, the
preparation process generates draft forms, annexures, and responses using approved evidence.
</text>
</box>
```

Stage 8 — Independent Verification Agent

```
text

  <text color="secondary" size="sm">A separate verification agent checks the draft package
against the original requirements and source evidence, rather than simply trusting the agents
that prepared it.</text>
</box>
```

Stage 9 — Verification decision

```
text

  <text color="secondary" size="sm">If critical checks fail, the workflow returns the failed
items for correction and re-verification. If they pass, the package moves forward. Unresolved
issues remain visible rather than being silently ignored.</text>
</box>
```

Stage 10 — Final deliverable

```
text
```

```
  <text color="secondary" size="sm">The user receives an approval-ready package containing draft bid documents, a compliance matrix, evidence references, a blocker report, and a record of verification status. Final approval and submission remain under human control.</text>
</box>
```

An example of how this works

Imagine a company applying for a government IT services tender. The tender requires:

- Minimum annual turnover of ₹5 crore.
- Three years of relevant project experience.
- A valid ISO certification.
- A signed declaration in the prescribed format.

The Evidence Agent finds audited financial statements, two project completion certificates, and an ISO certificate. The third project certificate is missing, and the declaration has not been signed.

TenderOps should not simply mark the tender as compliant.

Instead, it should produce a requirement-by-requirement assessment, flag the missing project evidence and signature, and ask the company to provide or resolve them. It can prepare the unsigned declaration as a draft and continue working on unrelated sections while waiting for the missing information.

Once the company supplies the required evidence, the relevant checks run again. The independent verifier then checks the completed package.

This is the core value proposition: **the system manages the entire process, including exceptions, instead of merely generating documents.**

---

5. The AI agents and their responsibilities

The system should use specialized agents with clearly defined responsibilities, rather than a collection of unrestricted LLMs that can all modify everything.

2. Evidence Agent

```
text

  <text color="secondary" size="sm">Primary responsibility: establish what the company can prove.</text>

  Searches company documents, retrieves relevant passages and records, checks document dates and metadata, and links evidence to tender requirements.

  <text size="sm" weight="medium">Output: Evidence-to-requirement mapping.</text>
```

```
</box>
```

writer pen and completed form" aspectRatio="1:1" maxWidth="88px"/> **4. Bid Preparation Agent**

```
text

  <text color="secondary" size="sm">Primary responsibility: prepare the response.</text>

  Drafts tender-specific answers, fills supported fields in templates, prepares annexures, and
assembles evidence-backed claims. It must not invent credentials or fill unknown fields with
guesses.

  <text size="sm" weight="medium">Output: Draft bid package and outstanding information
requests.</text>
</box>
```

5. Independent Verification Agent

```
text

  <text color="secondary" size="sm">Primary responsibility: catch errors before delivery.
</text>

  Re-examines mandatory requirements, checks source references, verifies numerical values,
detects unsupported claims and inconsistencies, and checks whether required documents are
present.

  <text size="sm" weight="medium">Output: Verification report with failed checks and
corrections required.</text>
</box>
```

6. Orchestrator

```
text

  <text color="secondary" size="sm">Primary responsibility: control execution.</text>

  Decides which agent runs next, maintains the shared workflow state, handles retries, routes
unresolved issues to humans, and determines whether the package is ready to progress.

  <text size="sm" weight="medium">Output: A coordinated, resumable execution of the full
workflow.</text>
</box>
```

The six responsibilities do not necessarily require six different models. The orchestrator and compliance engine can largely be implemented with ordinary code, while several agents can share the same underlying LLM with different instructions and tools.

That distinction is important for cost, latency, and reliability.

6. Handling failures and recovering from issues

Recovery is one of the most important aspects of TenderOps. Real tenders are messy: PDFs may be unreadable, company documents may be outdated, eligibility may be ambiguous, and agents may produce incomplete outputs.

The workflow therefore needs to distinguish between an execution error, a missing input, and a genuine compliance failure.

Situation	Appropriate behavior
A document cannot be parsed	Retry with an alternative parser or OCR; flag it if still unreadable.
A required certificate is missing	Request the certificate or mark the requirement unresolved.
Two documents contain conflicting figures	Preserve both sources and request reconciliation.
An LLM returns malformed output	Validate the schema and retry within a bounded limit.
An eligibility condition is ambiguous	Escalate for human interpretation rather than assume compliance.
A service times out	Retry safely, using idempotent operations where possible.
Verification finds an unsupported claim	Send the specific failure back for correction, then re-verify.

How recovery should work

flowchart TD A["Run agent task"] --> B{"Task successful?"} B -->|Yes| C["Validate output"] B -->|No| D{"Transient error?"} D -->|Yes| E["Bounded retry"] E --> A D -->|No| F["Save checkpoint and diagnose"] C --> G{"Requirements satisfied?"} G -->|Yes| H["Continue workflow"] G -->|No| I["Create blocker"] I --> J{"Can be resolved automatically?"} J -->|Yes| K["Correct or retrieve evidence"] K --> C J -->|No| L["Request human input"] L --> M["Resume from checkpoint"] M --> C

The critical design decision is that **the workflow should never treat failure to establish compliance as proof of compliance**.

For example, an unreadable certificate is not necessarily an invalid certificate, but it is also not sufficient evidence that the company qualifies. The system should preserve that uncertainty.

A checkpointed workflow makes this practical. If a human supplies a missing certificate, TenderOps should resume the affected tasks instead of reprocessing every document and repeating every agent call.

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7. Technical architecture

The proposed technical architecture separates workflow orchestration, document processing, evidence retrieval, compliance validation, and the user-facing application.

flowchart TB U["Web Application"] --> API["Backend API"] API --> STORE["Document Storage"] API --> INGEST["Document Ingestion Pipeline"] INGEST --> PARSE["PDF / DOCX / OCR Parsers"] PARSE --> REQ["Structured Requirements"] PARSE --> IDX["Search Index / Evidence Store"] REQ --> LG["LangGraph Orchestrator"] IDX --> EA["Evidence Agent"] LG --> TA["Tender Analyst Agent"] TA --> LG LG --> EA EA --> LG LG --> CE["Compliance Engine"] CE --> LG LG --> BA["Bid Preparation Agent"] BA --> LG LG --> VA["Independent Verification Agent"] VA --> LG LG --> HUMAN["Human Review / Blockers"] HUMAN --> LG LG --> OUT["Final Bid Package"] OUT --> STORE

Suggested technology stack

This is a proposed implementation, not a claim that these components are already integrated.

Component	Suggested technology	Purpose
Frontend	React + TypeScript	Tender workspace, status dashboard, evidence review and approvals
Backend API	FastAPI or Node.js	Authentication, uploads, workflow initiation and document delivery
Agent orchestration	LangGraph	State transitions, conditional routing, retries and resumable workflows
LLM layer	A capable instruction-following model	Clause interpretation, evidence analysis and document drafting
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Search and retrieval	PostgreSQL full-text search and/or vector search	Finding evidence in company documents
Structured data	PostgreSQL	Requirements, evidence links, statuses, audit events and workflow state
File storage	S3-compatible object storage	Original documents and generated artifacts

Component	Suggested technology	Purpose
Background p rocessing	A job queue and worker service	Long-running ingestion and bid-generati on tasks
Observability	neatlogs or equivalent tracing and logging tools	Execution traces, failures, latency, cost a nd agent evaluation

For an MVP, this can be simplified substantially. You do not need a vector database, multiple model providers, and a complicated microservice architecture from day one.

A PostgreSQL database with full-text search, object storage, a background worker, and LangGraph can cover a meaningful initial workflow. Add vector retrieval where semantic matching demonstrably improves evidence discovery.

The most important data model

TenderOps should treat requirements as structured records, not merely text buried in an LLM conversation.

A requirement record could contain:

- `requirement_id` : Unique identifier.
- `description` : What the tender demands.
- `category` : Eligibility, financial, technical, documentation, deadline, etc.
- `mandatory` : Whether the condition is compulsory.
- `source_reference` : Document, page, clause, and relevant excerpt.
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The evidence records should likewise preserve the source document, exact location, extraction confidence where available, relevant dates, and any human verification.

This creates an auditable relationship between the tender's requirements, the company's evidence, the compliance decisions, and the final bid.

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8. The human-in-the-loop model

TenderOps should automate work, not take unrestricted control of a company's bid.

Human review is particularly important for:

- Eligibility decisions involving ambiguous clauses.
- Conflicting financial or legal information.
- Missing mandatory documentation.
- Claims about past projects or company capabilities.
- Commercial commitments and pricing.
- Signing declarations and authorizing final submission.

A practical interface would show each requirement with its status, supporting evidence, source page, and next action.

Compliance review 2 unresolved Annual turnover Satisfied Audited financial statement · FY 2024–25 · Source reference attached Project experience Needs evidence Two relevant completion certificates found; tender requires three qualifying projects. Signed declaration Pending approval Draft generated. Authorized representative must review and sign.

GenUI.issueNewTurn("Design the detailed human-in-the-loop review interface and approval state machine for TenderOps, including permissions, audit logs, and unresolved compliance blockers.")}>Design the review interface

The ideal outcome is not that a human must review every piece of extracted text. It is that a human can focus on the small number of consequential decisions that the system cannot confidently resolve.

---

9. Reliability, observability, and the hackathon tooling

TenderOps is an agent-reliability problem as much as an AI application problem. A successful demonstration should show not only that the agents can produce a bid, but also that the workflow behaves predictably when things go wrong.

Three tools were included in the original workflow concept:

neatlogs — Runtime observability

```
text

  Instrument each workflow run to track agent inputs and outputs, tool calls, retries, latency,
token usage or cost where available, validation failures, and final outcomes.

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what was retried, and why the workflow stopped.
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Entire — Development and code provenance

```
text
```

```
  Use the development tool for coding-session checkpoints and change history where supported,
making it easier to understand how the implementation evolved and recover useful development
context.

  This is separate from production workflow checkpoints, which must be implemented in the
application itself.
</box>
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cfo.ai — Business and financial planning

```text
  Use the business-planning tool to examine the commercial viability of TenderOps: customer
pricing, operating costs, margins, customer acquisition, and potential runway.

  This is a business-planning layer, not part of the core tender-processing runtime.
</box>
```

How to demonstrate reliability

A strong hackathon demonstration would deliberately test failure cases:

1. Upload a normal tender and the company's supporting documents.
2. Introduce an unreadable PDF or malformed document.
3. Include an expired certificate or conflicting turnover figures.
4. Show the system detecting the problem rather than generating an unsupported compliance claim.
5. Demonstrate a saved checkpoint and a clear human-input request.
6. Supply the missing information and resume processing.
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This is a more compelling demonstration than showing only a successful happy path because it proves the system can manage exceptions.

---

10. What should the MVP include?

The biggest risk is trying to build a fully autonomous tendering platform immediately. A focused MVP should prove the complete workflow on one tender category before expanding.

Phase 1 — End-to-end proof of concept

```text
```

```
  - Upload one tender and a small company document bundle.
  - Extract and structure mandatory requirements.
  - Link requirements to source pages.
  - Retrieve supporting company evidence.
  - Produce a compliance matrix with explicit unresolved states.
  - Generate a basic bid checklist and missing-document report.
</box>
```

Phase 2 — Reliable workflow execution

```
text

  - Add LangGraph state management and checkpointing.
  - Add bounded retries, output validation and recovery paths.
  - Add the human-review queue.
  - Add independent verification.
  - Instrument workflow traces and build repeatable test cases.
</box>
```

Phase 3 — Bid generation

```
text

  - Populate supported fields in actual tender templates.
  - Generate draft declarations and technical responses.
  - Assemble annexures and evidence references.
  - Produce downloadable output documents.
  - Require human approval before finalization.
</box>
```

Phase 4 — Production readiness

```
text

  - Multi-user workspaces and access controls.
  - Secure company document repositories.
  - Versioned evidence and audit trails.
  - Better OCR and complex-table support.
  - Monitoring, evaluation datasets and operational reporting.
</box>
```

What I would measure

Metric	What it tells you
Requirement extraction r ecall	How many actual tender requirements the system identifies

Metric	What it tells you
Evidence retrieval precision	How often retrieved documents genuinely support the requirement
Compliance decision accuracy	Whether the system correctly distinguishes satisfied, failed, and unknown conditions
Unsupported-claim rate	How often generated documents contain claims without adequate evidence
Critical omission rate	How many mandatory requirements are missed in the final review
Recovery success rate	How often interrupted or failed workflows resume successfully
Human intervention rate	How frequently a workflow needs manual assistance
End-to-end processing time	How much time the system saves relative to a human baseline
Cost per tender	Whether the solution is commercially viable at the intended price

For a tendering product, false positives are especially dangerous: incorrectly claiming compliance with a mandatory condition could cause a company to lose a bid. Evaluation should therefore prioritize critical-requirement recall, grounded evidence, and conservative uncertainty handling, not just how polished the generated documents look.

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11. Business model and commercial opportunity

TenderOps could be sold as a B2B SaaS product, with pricing based on the volume and complexity of tenders processed.

Potential models include:

- **Subscription:** A monthly or annual plan for companies with regular tendering activity.
- **Usage-based pricing:** Charges based on the number of tenders, pages processed, or bid packages generated.
- **Enterprise plans:** Private document repositories, access controls, custom templates, audit capabilities, and integrations.
- **Assisted onboarding:** Initial setup of company credentials, document libraries, reusable evidence mappings, and standard bid templates.

The strongest long-term differentiator would be a reusable, verified company knowledge base. Once a company has organized its certifications, financial records, project references, and standard

declarations, the system can reuse that information across future tenders rather than beginning from scratch each time.

That accumulated evidence mapping could reduce preparation time and make TenderOps more useful with every subsequent bid.

The primary commercial challenge is proving that the system saves enough time without increasing the risk of missed requirements or incorrect submissions.

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12. What makes TenderOps different?

There are three possible layers of value.

First, document intelligence. The platform understands a tender and extracts actionable requirements.

Second, evidence-grounded execution. It checks whether the company can substantiate those requirements and creates draft responses tied to the underlying documents.

Third, reliable orchestration. It manages dependencies, interruptions, exceptions, retries, and independent verification across the entire process.

The third layer is particularly important. A generic LLM can summarize a tender or draft a response. TenderOps aims to manage the stateful process that connects the summary to actual evidence, completed documents, and a reviewable outcome.

The product should be judged by whether it can reliably complete that process, not by the number of agents it uses.

Final summary

TenderOps is best understood as an **AI operating system for tender participation**.

It accepts tender documents and company credentials, converts tender clauses into structured requirements, finds and validates supporting evidence, assesses eligibility, drafts the bid package, and independently checks the result. When it encounters missing documents, contradictions, ambiguous requirements, or execution failures, it records the issue, requests appropriate intervention, and resumes from a saved checkpoint.

The orchestrator coordinates the workflow, specialized agents handle individual tasks, deterministic code validates objective conditions, and humans retain control over consequential decisions.

For a hackathon, the strongest implementation would demonstrate one complete tender workflow with traceable evidence, a deliberately triggered failure, successful recovery, and an independent verification pass. That would establish the central thesis of TenderOps: **not just generating a bid with AI, but making the process of preparing a bid more reliable, auditable, and repeatable.**

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Human intervention rate	How frequently a workflow needs manual assistance
End-to-end processing ti me	How much time the system saves relative to a human baseline
Cost per tender	Whether the solution is commercially viable at the intended price

For a tendering product, false positives are especially dangerous: incorrectly claiming compliance with a mandatory condition could cause a company to lose a bid. Evaluation should therefore prioritize critical-requirement recall, grounded evidence, and conservative uncertainty handling, not just how polished the generated documents look.

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11. Business model and commercial opportunity

TenderOps could be sold as a B2B SaaS product, with pricing based on the volume and complexity of tenders processed.

Potential models include:

- **Subscription:** A monthly or annual plan for companies with regular tendering activity.
- **Usage-based pricing:** Charges based on the number of tenders, pages processed, or bid packages generated.
- **Enterprise plans:** Private document repositories, access controls, custom templates, audit capabilities, and integrations.
- **Assisted onboarding:** Initial setup of company credentials, document libraries, reusable evidence mappings, and standard bid templates.

The strongest long-term differentiator would be a reusable, verified company knowledge base. Once a company has organized its certifications, financial records, project references, and standard

declarations, the system can reuse that information across future tenders rather than beginning from scratch each time.

That accumulated evidence mapping could reduce preparation time and make TenderOps more useful with every subsequent bid.

The primary commercial challenge is proving that the system saves enough time without increasing the risk of missed requirements or incorrect submissions.

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12. What makes TenderOps different?

There are three possible layers of value.

First, document intelligence. The platform understands a tender and extracts actionable requirements.

Second, evidence-grounded execution. It checks whether the company can substantiate those requirements and creates draft responses tied to the underlying documents.

Third, reliable orchestration. It manages dependencies, interruptions, exceptions, retries, and independent verification across the entire process.

The third layer is particularly important. A generic LLM can summarize a tender or draft a response. TenderOps aims to manage the stateful process that connects the summary to actual evidence, completed documents, and a reviewable outcome.

The product should be judged by whether it can reliably complete that process, not by the number of agents it uses.

Final summary

TenderOps is best understood as an **AI operating system for tender participation**.

It accepts tender documents and company credentials, converts tender clauses into structured requirements, finds and validates supporting evidence, assesses eligibility, drafts the bid package, and independently checks the result. When it encounters missing documents, contradictions, ambiguous requirements, or execution failures, it records the issue, requests appropriate intervention, and resumes from a saved checkpoint.

The orchestrator coordinates the workflow, specialized agents handle individual tasks, deterministic code validates objective conditions, and humans retain control over consequential decisions.

For a hackathon, the strongest implementation would demonstrate one complete tender workflow with traceable evidence, a deliberately triggered failure, successful recovery, and an independent verification pass. That would establish the central thesis of TenderOps: **not just generating a bid with AI, but making the process of preparing a bid more reliable, auditable, and repeatable.**